

Project White Paper — Java Code ReStructor

Project Goal

Develop a **cross-platform, embeddable Java code formatter** that:

-  Sorts class members (constants, fields, constructors, methods, etc.) according to a defined structure.
-  Formats code using an external formatter (e.g., Palantir Java Format).
-  Works independently of any IDE, CI platform, or editor.
-  Can be embedded as a CLI tool, Maven/Gradle plugin, Docker job, etc.

Why Existing Solutions Don't Work

JetBrains IDEA Code Style

- Works only inside IntelliJ IDEA.
- Requires manual formatting by the developer.
- Cannot be enforced via CI.
- Configuration is not portable across environments.

Qodana

- Can validate member order only in the paid Ultimate edition.
- Does not auto-correct or sort code — just flags violations.
- Requires Docker, token management, and separate pipeline configuration.
- Not suitable for local development; overkill for simple CI enforcement.

Google Java Format / Palantir Java Format

- Great for formatting code layout (indentation, spacing).
-  Do not sort class members by structure or visibility.

Spotless + Checkstyle

- Allow partial rule enforcement via custom checks.
- Do not perform actual reordering of code.
- Complex to configure, not scalable across teams.

What Our Tool Will Do

Functionality

- Parses `.java` source files into an **AST** (Abstract Syntax Tree) using `JavaParser` or a similar library.
- Builds a **dependency graph for constants**, ensuring correct declaration order.
- Sorts class members based on configuration:
 - Grouping: `static final` constants, fields, constructors, getters/setters, methods.
 - Sorting inside groups: by **visibility** (public → private), optionally by type, name, annotations.
- Applies external formatter (e.g., **Palantir Java Format**) after sorting.

Supported Modes

- `sort` : reorder and rewrite files.

- `check` : validate correct order (for CI).
- `format` : apply external formatter only.
- `fix` : full sort + format in one pass.

⚙️ Configuration Sources

- Built-in default config.
- Exported IDE configs (e.g., `.editorconfig` , XML).
- Custom YAML config file (e.g., `sorter-config.yml`).
- Inline config via CLI or plugin params.

📦 Integration Options

- CLI tool (local & CI use).
- Maven/Gradle plugin.
- Docker image.
- Optional: IDE plugin in the future.

✅ Why This Matters

- 🖥️ **IDE-independent**: works with any editor or environment.
- 🔒 Enables CI Quality Gates (e.g., GitLab).
- 📄 Reduces noisy diffs, improves PR readability.
- ⚙️ Fully automatable — no manual effort.
- 🌍 Can be released as open-source to represent **UBS engineering excellence**.